CHAIN RULE

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Name: Solutions

Theorem (The Chain Rule)**.** *Assume that g is differentiable at x and f is differentiable at $g(x)$. Then composition of f with g, $f \circ g(x) = f\big(g(x)\big)$, is differentiable at x and*

$$\frac{\mathrm{d}}{\mathrm{d}x} f \circ g(x) = f'\big(g(x)\big) \cdot g'(x).$$

1. Let $f(x) = (3x^2 + 1)^2$.

(a) Expand $f(x)$, then take the derivative.

$$\frac{d}{dx}\left(9x^4 + 6x^2 + 1\right) = 36x^3 + 12x$$

(b) Write $f(x) = (3x^2 + 1)^2 = (3x^2 + 1)(3x^2 + 1)$ and apply the Product Rule.

$$f'(x) = 6x(3x^2+1) + (3x^2+1)(6x) = 2(6x)(3x^2+1)$$
$$= 12x(3x^2+1) = 36x^3 + 12x$$

(c) Apply the chain rule directly to $f(x)$.

$$f'(x) = 2(3x^2+1)(6x) = 36x^3 + 12x$$

(d) Are your answers in parts (a), (b), and (c) the same? Why or why not?

Yes; these are all valid ways to compute the same thing: the derivative of f.

2. Assume that f is a differentiable function and let $g(x) = \left(f\left(\sqrt{x}\right)\right)^3$.

(a) Compute $g'(x)$. Your answer should include both f and f'.

$$g'(x) = \boxed{\frac{3f(\sqrt{x})^2 f'(\sqrt{x})}{2\sqrt{x}}}$$

(b) If $f(2) = 1$ and $f'(2) = -2$, calculate $g'(4)$.

$$g'(4) = \frac{3(1)^2(-2)}{4} = \frac{-6}{4} = \boxed{-\frac{3}{2}}$$

Find the derivative of the given function.

3. $f(x) = (1 + x + x^2)^{99}$

$$f'(x) = \boxed{99(1+x+x^2)^{98}(2x+1)}$$

4. $g(\theta) = (2 - \sin(\theta))^{3/2}$

$$g'(\theta) = \tfrac{3}{2}\left(2-\sin(\theta)\right)^{1/2}\left(-\cos(\theta)\right)$$

$$= \boxed{\frac{-3\cos(\theta)\sqrt{2-\sin(\theta)}}{2}}$$

5. $f(x) = (2x - 3)^4(x^2 + x + 1)^5$.

$$f'(x) = 4(2x-3)^3(2)(x^2+x+1)^5 + (2x-3)^4(5)(x^2+x+1)^4(2x+1)$$

$$= (2x-3)^3(x^2+x+1)^4\left[8(x^2+x+1) + 5(2x+1)(2x-3)\right]$$

$$= (2x-3)^3(x^2+x+1)^4\left[8x^2+8x+8 + 20x^2-20x-15\right]$$

$$= \boxed{(2x-3)^3(x^2+x+1)^4(28x^2-12x-7)}$$

6. $g(\theta) = \cos(\theta)^2$

$$g'(\theta) = 2\cos(\theta)(-\sin(\theta)) = \boxed{-2\cos(\theta)\sin(\theta)}$$

7. $h(y) = \left(\dfrac{y^4+1}{y^2+1}\right)^5$

$$h'(y) = 5\left(\frac{y^4+1}{y^2+1}\right)^4\left(\frac{4y^3(y^2+1) - (y^4+1)(2y)}{(y^2+1)^2}\right)$$

$$= \frac{5(y^4+1)^4\left(4y^5+4y^3-2y^5-2y\right)}{(y^2+1)^6}$$

$$= \boxed{\frac{5(y^4+1)(2y^5+4y^3-2y)}{(y^2+1)^6}}$$

8. $f(\theta) = \cot^2(\sin(\theta))$

$$f'(\theta) = 2\cot(\sin(\theta))\left(-\csc^2(\sin(\theta))\right)\cos(\theta)$$

$$= \boxed{-2\cot(\sin(\theta))\csc^2(\sin(\theta))\cos(\theta)}$$

9. $f(t) = \sqrt{\dfrac{t}{t^2+4}}$

$$f'(t) = \frac{1}{2}\left(\frac{t}{t^2+4}\right)^{-1/2}\left(\frac{t^2+4-t(2t)}{(t^2+4)^2}\right)$$

$$= \frac{1}{2}\,\frac{\sqrt{t^2+4}}{\sqrt{t}}\left(\frac{-t^2+4}{(t^2+4)^2}\right)$$

$$= \boxed{\frac{-t^2+4}{2\sqrt{t}\,(t^2+4)^{3/2}}}$$